

Pattern formation in a minimal cross-feeding model: steady states and linear stability

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Abstract

A population consumes an external resource, leaks a fixed fraction of the consumed flux as a byproduct, and re-consumes that byproduct. We characterise the homogeneous steady states (existence, positivity, stability) and then carry out a linear stability analysis in space under a quasi-steady-state assumption for both resources. The dispersion relation is an activator–inhibitor form with one gain term and two loss terms, the second of which is doubly screened. We show that the unstable set is always a single connected band, and reduce the instability criterion to a closed-form window $\beta_{\min} \leq \beta < \beta_c(p, \gamma, l)$ together with an upper bound on p . That bound is piecewise: which branch applies is decided by $l(1 + \gamma)$ against 1, and only on the upper branch can p exceed unity.

1 Model and notation

The model is

$$\partial_t N = N(c(1-l)I + dR - m) + D_N \nabla^2 N, \quad (1)$$

$$\partial_t I = K - rI - cNI + D_I \nabla^2 I, \quad (2)$$

$$\partial_t R = clNI - rR - dNR + D_R \nabla^2 R, \quad (3)$$

with N the population, I the externally supplied resource and R the leaked byproduct. Cells consume I at rate cNI ; a fraction l of that flux is leaked as R and the remaining fraction $1-l$ becomes growth. The byproduct is re-consumed by the same population at rate dNR , also contributing to growth. Both resources are supplied or removed at dilution rate r , and K is the external supply.

Throughout we use

$$\gamma \equiv \frac{d}{c}, \quad p \equiv \frac{D_R}{D_I}, \quad \beta \equiv \frac{Kc}{mr}, \quad (4)$$

so $d = \gamma c$ and $D_R = pD_I$. Here γ compares the two uptake rates, p the two resource diffusivities, and β is a supply-to-maintenance ratio: K/r is the resource level in the absence of cells, cK/r the resulting per-capita gross growth rate, and β measures that against the death rate m .

Two combinations recur constantly and are worth naming immediately:

$$u \equiv r + cN, \quad v \equiv r + dN. \quad (5)$$

Each is a total removal rate — u is the rate at which I leaves the system (dilution plus consumption) and v likewise for R .

2 Homogeneous steady states

2.1 The steady state condition

Setting all time derivatives and Laplacians to zero, (2) and (3) give

$$I^* = \frac{K}{u}, \quad R^* = \frac{clKN^*}{uv}, \quad (6)$$

with u, v evaluated at N^* . Equation (1) with $N^* \neq 0$ requires the bracket to vanish,

$$G(N^*) \equiv c(1-l)I^* + dR^* - m = 0. \quad (7)$$

Writing $G(N) = -F(N)/(uv)$ with

$$F(N) = mcdN^2 + (mr(c+d) - Kcd)N + mr^2 - Kcr(1-l), \quad (8)$$

the positive steady states are the positive roots of the quadratic F . In terms of the scaled variable cN/r this is

$$\gamma\left(\frac{cN}{r}\right)^2 + \underbrace{(1+\gamma-\beta\gamma)}_{b/r} \frac{cN}{r} + (1-\beta(1-l)) = 0. \quad (9)$$

2.2 The reduced variable V

It is convenient to work with the scaled byproduct removal rate

$$V \equiv \frac{v}{r} = 1 + \gamma \frac{cN}{r} \geq 1, \quad (10)$$

in terms of which $u/r = (V + \gamma - 1)/\gamma$ and (9) becomes the monic quadratic

$$\boxed{V^2 - (1 + \beta\gamma - \gamma)V + \beta\gamma l = 0} \quad (11)$$

(this is γ^{-1} times (9) after substitution). The single most useful consequence is that (11) can be solved for β explicitly:

$$\beta(V) = \frac{V(V + \gamma - 1)}{\gamma(V - l)}. \quad (12)$$

So rather than solving a quadratic for V and substituting, one can parametrise the steady-state branch by V and read off the corresponding β . Every condition below is first derived as an inequality on V and then converted through (12).

Differentiating (12),

$$\frac{d\beta}{dV} \propto V^2 - 2lV - (\gamma - 1)l, \quad (13)$$

whose positive root is exactly the fold identified in Section 2.3. Hence β is strictly increasing along the upper branch, and inequalities may be transferred between V and β without changing direction.

2.3 Existence and positivity

Two separate questions arise: when are the roots of (11) real, and when do they correspond to $N > 0$, i.e. $V > 1$.

Reality. The discriminant of (9) is

$$\mathcal{D} = r^2 \left[\gamma^2 \beta^2 + 2\gamma(1 - \gamma - 2l)\beta + (1 - \gamma)^2 \right], \quad (14)$$

a quadratic in β whose own discriminant is

$$\mathcal{D}' = 16\gamma^2 l (l + \gamma - 1). \quad (15)$$

If $l + \gamma < 1$ then $\mathcal{D}' < 0$ and the roots are real for every β . Otherwise $\mathcal{D}' < 0$ precisely for β strictly between

$$\beta_{\pm} = \frac{2l + \gamma - 1 \pm 2\sqrt{l(l + \gamma - 1)}}{\gamma} = \frac{(\sqrt{l} \pm \sqrt{l + \gamma - 1})^2}{\gamma}. \quad (16)$$

Positivity. Evaluating the left-hand side of (11) at $V = 1$ gives γ^{-1} times $\gamma[1 - \beta(1 - l)]$, so

$$\beta > \frac{1}{1 - l} \iff \text{exactly one root has } V > 1. \quad (17)$$

Below that threshold the two roots lie on the same side of 1, and both exceed 1 only if the vertex does, i.e. if $\beta > (1 + \gamma)/\gamma$.

Proposition 1 (Ordering). *If $l(1 + \gamma) > 1$ then*

$$\beta_- < \frac{1 + \gamma}{\gamma} < \beta_+ \leq \frac{1}{1 - l}. \quad (18)$$

If $l(1 + \gamma) < 1$ the interval $((1 + \gamma)/\gamma, 1/(1 - l))$ is empty, and no positive pair exists below the transcritical threshold.

Proof. At $\beta = (1 + \gamma)/\gamma$ the linear coefficient b of (9) vanishes, so there $\mathcal{D} = -4\gamma r^2(1 - \beta(1 - l))$, which is negative exactly when $\beta < 1/(1 - l)$, i.e. when $l(1 + \gamma) > 1$. A negative discriminant means $(1 + \gamma)/\gamma$ lies strictly between β_- and β_+ , giving the first two inequalities. At $\beta = 1/(1 - l)$ the constant term vanishes and $\mathcal{D} = b^2 \geq 0$, so $1/(1 - l)$ never lies strictly inside (β_-, β_+) ; since it exceeds $(1 + \gamma)/\gamma$, which is inside, it must be at least β_+ . The second statement is the reverse inequality $(1 + \gamma)/\gamma > 1/(1 - l)$. $\square$

Proposition 1 disposes of the branch $\beta < \beta_-$: it lies below $(1 + \gamma)/\gamma$, where the vertex of (9) is at negative N , so both roots are negative and no positive steady state exists there. Only $\beta > \beta_+$ is relevant.

Existence threshold. Collecting the two mechanisms — emergence at a fold versus emergence transcritically from $N = 0$ — the lowest β admitting a positive steady state is

$$\beta_{\min} = \begin{cases} \frac{1}{1 - l}, & l(1 + \gamma) \leq 1 \quad (\text{regime A}), \\ \frac{(\sqrt{l} + \sqrt{l + \gamma - 1})^2}{\gamma} = \beta_+, & l(1 + \gamma) > 1 \quad (\text{regime B}). \end{cases} \quad (19)$$

The two branches agree at $l(1 + \gamma) = 1$, so $\beta_{\min}$ is continuous with a kink there. The corresponding value of V at onset is

$$V_{\min} = \begin{cases} 1, & \text{regime A,} \\ \sqrt{l}(\sqrt{l} + \sqrt{l + \gamma - 1}), & \text{regime B,} \end{cases} \quad (20)$$

the regime-B value being the double root $(1 + \beta_+\gamma - \gamma)/2$ of (11). Note that $V_{\min}$ is precisely the positive root of (13), confirming the monotonicity claim made there.

Range of β	States	Stability
$\beta < \beta_{\min}$	$N = 0$ only	extinction
$\beta_+ \leq \beta < \frac{1}{1-l}$ (regime B only)	$0, N_-, N_+$	bistable: 0 and N_+ stable, N_- unstable
$\beta > \frac{1}{1-l}$	$0, N_+$	N_+ stable, 0 unstable

Table 1: Well-mixed phase structure. In regime A the middle row is empty and the positive state appears transcritically at $\beta = 1/(1-l)$.

2.4 Stability without space

Since $G(N) = -F(N)/(uv)$ and $G(N^*) = 0$, the well-mixed eigenvalue is

$$\lambda_0 = N^* G'(N^*) = -\frac{N^* F'(N^*)}{uv}, \quad (21)$$

so a positive steady state is stable iff $F'(N^*) > 0$, i.e. iff it is the *larger* root. The extinction state $N = 0$ has eigenvalue $G(0) = c(1-l)K/r - m$, so it is stable iff $\beta(1-l) < 1$. The resulting picture is summarised in Table 1.

Remark. In regime B the pattern-forming state coexists with a stable extinction state. The linear analysis below says nothing about which basin the system occupies; that must be settled separately.

3 Linear stability in space

3.1 Setup and the quasi-steady-state reduction

Perturb about the homogeneous state,

$$N = N^* + \nu e^{i\mathbf{k}\cdot\mathbf{x} + \lambda t}, \quad I = I^* + \rho e^{i\mathbf{k}\cdot\mathbf{x} + \lambda t}, \quad R = R^* + \sigma e^{i\mathbf{k}\cdot\mathbf{x} + \lambda t}, \quad (22)$$

so that each Laplacian becomes multiplication by $-k^2$. We assume the resources equilibrate fast and set $\partial_t \rho = \partial_t \sigma = 0$ while retaining their diffusion. This reduces a 3×3 eigenvalue problem to a scalar equation.

Linearising (2) gives

$$\rho = -\frac{c I^*}{u + D_I k^2} \nu, \quad (23)$$

which has the opposite sign to ν : more cells means less external resource. This is the *inhibitory* channel, and it is suppressed at large k because rapid diffusion smooths out any local depletion.

Linearising (3), and using $c l I^* - d R^* = (c l K/u)(r/v)$, which follows from (6) and $v - d N^* = r$,

$$\sigma = \frac{1}{v + D_R k^2} \left[\frac{c l K r}{uv} \nu + c l N^* \rho \right]. \quad (24)$$

The first term is positive — more cells leak more byproduct, the *activating* channel — while the second inherits the negative ρ and is a second inhibitory contribution, this one routed through the byproduct.

Finally, in (1) the perturbation ν multiplies the bracket, which vanishes at the fixed point by (7). Only N^* times the bracket's perturbation survives:

$$\lambda \nu = N^* [c(1-l)\rho + d\sigma] - D_N k^2 \nu. \quad (25)$$

This is the key structural fact: N has no autonomous linear dynamics, and its entire growth rate is inherited from the resource perturbations.

3.2 The dispersion relation

Substituting (23) and (24) into (25) and dividing by ν , then factoring each denominator using the screening lengths

$$\ell_I = \sqrt{\frac{D_I}{u}}, \quad \ell_R = \sqrt{\frac{D_R}{v}}, \quad (26)$$

gives

$$\lambda(k) = -D_N k^2 + \frac{\mathcal{A}}{1 + \ell_R^2 k^2} - \frac{\mathcal{I}_1}{1 + \ell_I^2 k^2} - \frac{\mathcal{I}_2}{(1 + \ell_I^2 k^2)(1 + \ell_R^2 k^2)} \quad (27)$$

with all three coefficients positive:

$$\mathcal{A} = \frac{N^* d c l K r}{u v^2}, \quad \mathcal{I}_1 = \frac{N^* c^2 (1 - l) K}{u^2}, \quad \mathcal{I}_2 = \frac{N^{*2} d c^2 l K}{u^2 v}. \quad (28)$$

(The calligraphic $\mathcal{I}_1, \mathcal{I}_2$ are loss coefficients and are not to be confused with the resource I .) This is an activator–inhibitor form: one gain term, two loss terms.

Checks. At $k = 0$ all three fractions equal unity and $\lambda(0) = \mathcal{A} - \mathcal{I}_1 - \mathcal{I}_2$, which equals $N^* G'(N^*)$, the well-mixed eigenvalue; on the stable branch it is negative. At $k \rightarrow \infty$ with $D_N = 0$ every term vanishes and $\lambda \rightarrow 0$, which is the consequence of (25): screen out the resources and N has nothing left to do.

Why $\mathcal{I}_2$ is different. At large k the first two terms decay as k^{-2} while $\mathcal{I}_2$ decays as k^{-4} , because that inhibitory signal passes through two screened resource fields in sequence. Its second moment in real space corresponds to a width $\sqrt{\ell_I^2 + \ell_R^2}$, longer than the activator's ℓ_R for *any* diffusivities. The separation of scales here therefore comes from path length through the metabolic chain rather than from a diffusivity ratio. (Its exponential tail, by contrast, is set by $\max(\ell_I, \ell_R)$, so the statement is one about widths, not tails.)

3.3 How many unstable bands?

Both Lorentzian denominators are strictly positive, so multiplying through by them preserves the sign of λ . With $s = k^2$ define

$$P(s) \equiv (1 + \ell_I^2 s)(1 + \ell_R^2 s)\lambda(s), \quad \text{sign } \lambda = \text{sign } P. \quad (29)$$

Proposition 2 ($D_N = 0$: at most one crossing). *For $D_N = 0$,*

$$P(s) = \underbrace{(\mathcal{A} - \mathcal{I}_1 - \mathcal{I}_2)}_{=\lambda(0)<0} + (\mathcal{A}\ell_I^2 - \mathcal{I}_1\ell_R^2)s, \quad (30)$$

which is linear in s . Hence λ changes sign at most once, and does so iff $\mathcal{A}\ell_I^2 > \mathcal{I}_1\ell_R^2$, at $s_ = (\mathcal{I}_1 + \mathcal{I}_2 - \mathcal{A})/(\mathcal{A}\ell_I^2 - \mathcal{I}_1\ell_R^2)$.*

The cancellation is exact because $\mathcal{A}$ picks up only the factor $(1 + \ell_I^2 s)$, $\mathcal{I}_1$ only $(1 + \ell_R^2 s)$, and $\mathcal{I}_2$ — being doubly screened — picks up neither. Two consequences follow. First, the large- k balance is not merely an asymptotic heuristic: it decides the sign of λ everywhere, so the condition $\mathcal{A}\ell_I^2 > \mathcal{I}_1\ell_R^2$ is necessary as well as sufficient. Second, $\mathcal{I}_2$ never appears in the slope, so it can only shift the band edge s_* , never open or close a band.

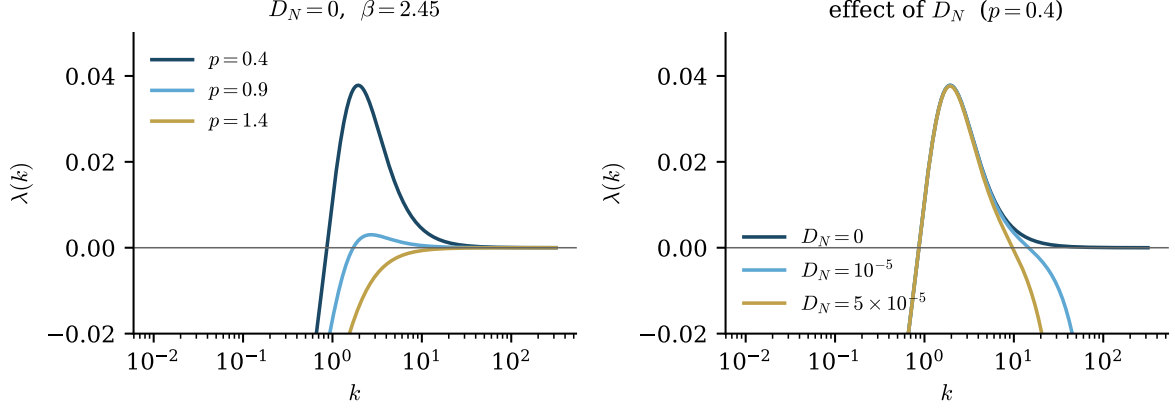


Figure 1: Dispersion relations from (27) with $c = r = m = 1$, $\gamma = 1$, $l = 0.6$, $D_I = 1$. *Left:* $D_N = 0$; decreasing p moves the system across the threshold, and once unstable the band extends to $k \rightarrow \infty$. *Right:* nonzero D_N closes the band from above, selecting a finite wavelength; large enough D_N suppresses it entirely.

Proposition 3 ($D_N > 0$: a single connected band). *For $D_N > 0$,*

$$P(s) = -D_N \ell_I^2 \ell_R^2 s^3 - D_N (\ell_I^2 + \ell_R^2) s^2 + c_1 s + (\mathcal{A} - \mathcal{I}_1 - \mathcal{I}_2), \quad c_1 \equiv \mathcal{A} \ell_I^2 - \mathcal{I}_1 \ell_R^2 - D_N, \quad (31)$$

and $P'(s) = -3D_N \ell_I^2 \ell_R^2 s^2 - 2D_N (\ell_I^2 + \ell_R^2) s + c_1$ has exactly one positive root when $c_1 > 0$ and none when $c_1 \leq 0$. Therefore P either decreases monotonically on $s > 0$, or rises once and then falls. Starting from $P(0) < 0$, it has either no positive roots or exactly two: the unstable set is always a single interval $[k_-, k_+]$, never two disjoint bands.

In particular $c_1 > 0$, that is

$$D_N < \mathcal{A} \ell_I^2 - \mathcal{I}_1 \ell_R^2, \quad (32)$$

is necessary for instability, though not sufficient — the band must also actually open, which requires $P > 0$ at the unique turning point. As $D_N \rightarrow 0^+$ the upper edge $k_+ \rightarrow \infty$ and the half-line of the previous proposition is recovered. Uniqueness of the band also means the fastest-growing mode is unique, so wavelength selection is well posed.

3.4 The instability condition

Write out $\mathcal{A} \ell_I^2 > \mathcal{I}_1 \ell_R^2$ using (28). The common factor $N^* c K / u^2$ cancels from both sides, leaving $d l r D_I / v^2 > c(1 - l) D_R / v$, i.e.

$$d l r D_I > c(1 - l) D_R v. \quad (33)$$

Substituting $d = \gamma c$, $D_R = p D_I$ and dividing by $c r D_I$ gives a bound on V alone:

$$V < V_c \equiv \frac{\gamma}{p} \cdot \frac{l}{1 - l} \quad (34)$$

Every parameter enters only through this comparison. The interpretation is direct: $V = v/r$ measures how fast the byproduct is removed relative to dilution, and a large V means the byproduct is consumed before it can diffuse anywhere, destroying the activation. Patterning therefore requires a *sparse* population.

Converting (34) through $\beta(V)$ from (12), and using the monotonicity established after (13), gives the upper edge of the window in closed form:

$$\boxed{\beta_c = \beta(V_c) = \frac{\gamma l + p(\gamma - 1)(1 - l)}{p(1 - l)(\gamma - p(1 - l))}} \quad (35)$$

At $p = 1$ the numerator reduces to $\gamma - 1 + l$ and the denominator to $(1 - l)(\gamma - 1 + l)$, so $\beta_c = 1/(1 - l)$: with equal diffusivities the condition is exactly $\beta < 1/(1 - l)$, i.e. the population must be unable to survive on the external resource alone and must be propped up by its own byproduct.

3.5 The bound on p

The window $\beta_{\min} \leq \beta < \beta_c$ is non-empty iff $V_c > V_{\min}$, with $V_{\min}$ from (20). This is also the condition under which (35) is meaningful at all: $\beta(V)$ is monotonic only above the fold, so evaluating it at a V_c below $V_{\min}$ returns a number that is not a threshold. The primitive statement is always (34). Requiring $V_c > V_{\min}$ gives the piecewise bound

$$\boxed{p < \frac{\gamma l}{(1 - l)V_{\min}} = \begin{cases} \frac{\gamma l}{1 - l}, & l(1 + \gamma) \leq 1 \quad (\text{regime A}), \\ \frac{\gamma \sqrt{l}}{(1 - l)(\sqrt{l} + \sqrt{l + \gamma - 1})}, & l(1 + \gamma) > 1 \quad (\text{regime B}). \end{cases}} \quad (36)$$

The distinction matters. In regime A the condition $l(1 + \gamma) \leq 1$ rearranges to $\gamma l/(1 - l) \leq 1$, so the bound on p can never exceed unity: with a transcritical onset the byproduct must diffuse no faster than the external resource. Only in regime B, where the state is born at a fold, can $p > 1$ be admissible. Using the regime-A expression outside its range overestimates the bound; at $\gamma = 1$, $l = 0.6$ it gives $p < 1.5$ where the correct value is $p < 1.25$.

At $\gamma = 1$ the regimes divide at $l = \frac{1}{2}$ and (36) reads

$$p < \frac{l}{1 - l} \quad (l \leq \tfrac{1}{2}), \quad p < \frac{1}{2(1 - l)} \quad (l > \tfrac{1}{2}), \quad (37)$$

the two meeting at $p < 1$ when $l = \frac{1}{2}$.

4 Summary

For $D_N = 0$, the homogeneous state is unstable to spatial perturbations iff

$$p < \frac{\gamma l}{(1 - l)V_{\min}} \quad \text{and} \quad \beta_{\min} \leq \beta < \beta_c, \quad (38)$$

with $V_{\min}$, $\beta_{\min}$, β_c given by (20), (19), (35). The unstable set is then a single interval extending to $k \rightarrow \infty$. For $D_N > 0$ the band is finite and a wavelength is selected; (32) is necessary, and sufficiency requires $P > 0$ at its turning point. If $D_I = D_R = 0$ then λ is k -independent and no instability is possible at any k .

Three ingredients are therefore distinct and all required: a sign condition (34) that is mostly a statement about reaction kinetics; a screening condition, at least one $D > 0$, so that the sign flip actually happens as k grows; and a smallness condition on D_N , which alone damps rather than merely screening.

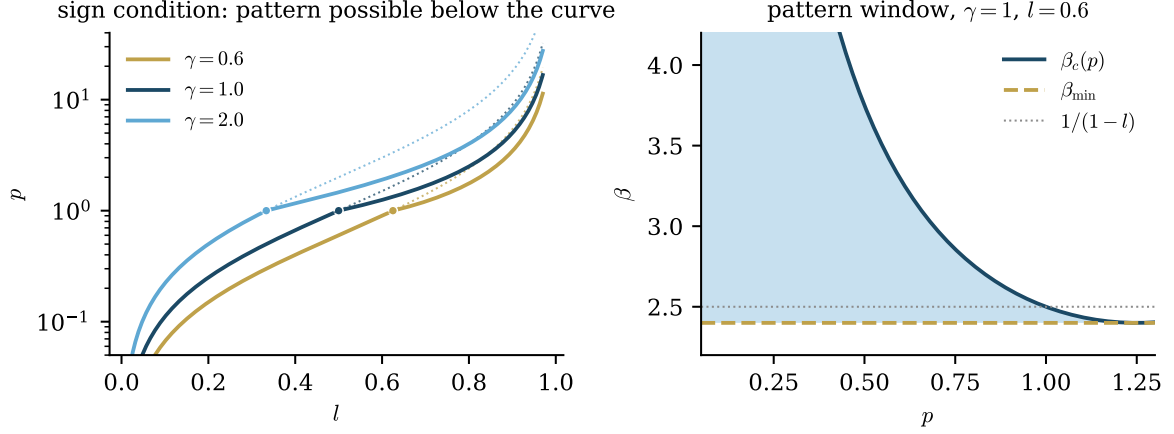


Figure 2: *Left*: the bound (36) in the (l, p) plane for three values of γ ; patterning is possible below each curve. Dotted lines continue the regime-A expression beyond its range of validity, and markers show the kink at $l(1 + \gamma) = 1$, which is also where the curve crosses $p = 1$. *Right*: the window $\beta_{\min} \leq \beta < \beta_c(p)$ at $\gamma = 1, l = 0.6$, closing at $p = 1.25$.

A Numerical verification

The following were checked against direct numerics over several hundred random parameter draws spanning several decades in each rate and diffusivity: the dispersion relation (27) against a direct solve of the quasi-steady-state linear system (agreement to 2×10^{-14} relative); the identity $\lambda(0) = N^* G'(N^*)$ against finite differences of G ; the equivalence of $\mathcal{A}\ell_I^2 > \mathcal{I}_1\ell_R^2$, of $V < V_c$, and of the numerically observed sign of $\max_k \lambda$ (no mismatches); the closed form (35) against $\beta(V_c)$; the piecewise $\beta_{\min}$ (19) against a bisection for the lowest β admitting a positive stable state; the ordering of Proposition 1; the identification of $V_{\min}$ with the root of (13); and the claim that λ never changes sign more than twice when $D_N > 0$.